

# Show your work

Exam FINAL Dec 9<sup>th</sup> 2025 Chem 1127

Name (First, Last):

Key

pg 1-6: questions, pg 7-9: Eqns, P.T. & spare pg.

Total Points: \_\_\_\_ + 2SET = \_\_\_\_ / 120<sub>124</sub>

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Show your work

1. An unknown element is a mixture of isotopes  $^{129}\text{X}$  and  $^{132}\text{X}$ . The average atomic mass of X is 129.5 amu. What is the percent abundance of  $^{129}\text{X}$ ? (3 pt)

No work  
-2.5

83.3%

$$Y + Z = 1 \rightarrow Y = 1 - Z$$

$$129 \cdot Y + 132 \cdot Z = 129.5$$

$$129(1 - Z) + 132Z = 129.5$$

$$129 - 129Z + 132Z = 129.5$$

$$3Z = 0.5$$

$$Z = 0.167 \rightarrow 16.7\%$$

if ans  
-1.5

- a. If there are only two isotopes of X, what must their % abundances add up to equal? (1 pt)

100%

83.3%

2. For the  $n = 4$  level/shell of an atom, clearly identify all possible values for  $\ell$ , and for each  $\ell$  clearly identify all possible values of  $m_\ell$ . (2 pt)

$n = 4$

$\ell = 0$        $\ell = 1$        $\ell = 2$        $\ell = 3$

$m_\ell = 0$        $m_\ell = -1, 0, 1$        $m_\ell = -2, -1, 0, 1, 2$        $m_\ell = -3, -2, -1, 0, 1, 2, 3$

- a. Identify a number for  $\ell$ , and identify the letter associated with it, and how many orbitals are in this sublevel/subshell. (1 pt)  $\ell = \underline{0}$  letter of  $\ell = \underline{s}$  # of orbitals = 1

2 - d - 5  
3 - f - 7

2. For the three quantum numbers ( $n, \ell, m_\ell, m_s$ ) match the symbol with the associated trait. (2 pt)

- a. Shape:  $\ell$  b. Spin:  $m_s$  c. Energy/Size:  $n$  d. Direction/Orientation:  $m_\ell$

3. Which orbital has the shape of a sphere? Circle it. (1 pt)

a. s b. f c. p d. d

- a. Briefly define the shape of a p orbital. Hourglass / Dumb bell (1 pt)

4. An ion,  $^{80}\text{Br}^-$ , contains how many protons (p), neutrons (n), and electrons (e) and identify this as an anion or cation. (2 pt)

$Z = 35$        $80 - 35 = 45$       36

anion or cation.

5. Circle the below element(s) that is (are) diatomic in their elemental form. (2 pt)

H

F

N

P

6. For the following blanks fill them in with proton (p), electron (e), or neutron (n). Note, one or more of the blanks should be left blank. (3 pt)

- a. p & n is (are) the heaviest or of equally light mass.

- b. n & e is (are) neutral.

- c. e &    reside outside the nucleus of an atom.

7. If the energy of electromagnetic radiation decreases, how does the wavelength & frequency change? Do they stay constant, increase, or decrease? (2 pt)

$E = h \cdot \nu$   
 $c = \nu \cdot \lambda$

wavelength

frequency

Flipped -1.5

8. Clearly identify how many significant figures are in the values below and circle the one which is given in scientific notation. (2 pt)

a.  $0.45 \times 10^{-5}$

b. 0.02504 4

c.  $-2.60 \times 10^{-4}$

3

9. Under standard conditions, what is the density (in g/mL) of water. (1 pt) a. 100 b. 1 c. 0.0001 d. 0

Key

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h

Key

10. A photon of light has frequency of 450 MHz, what is its wavelength in meters? Provide your answer with the correct number of significant figures. (2 pt)

3 sig figs  
100%

$$\nu = \frac{450 \text{ MHz}}{1} \cdot \frac{10^6 \text{ Hz}}{1 \text{ MHz}} = 4.5 \cdot 10^8 \text{ Hz or } \frac{1}{s}$$

$$c = \nu \cdot \lambda \rightarrow \lambda = \frac{c}{\nu} = \frac{3 \cdot 10^8 \text{ m/s}}{4.5 \cdot 10^8 \text{ Hz}} = 0.67 \text{ m}$$

- a. How can the units of Hz be rewritten in terms of seconds (s)? (1 pt)  $\frac{1}{s}$

- b. What is the energy of 1.3 moles of the above photons. (1.5 pt)

$$E = h \cdot \nu = 6.63 \cdot 10^{-34} \text{ J} \cdot \text{s} \cdot 4.5 \cdot 10^8 \frac{1}{s} = E = 2.984 \cdot 10^{-25} \text{ J/photon}$$

$$\frac{1.3 \text{ mol}}{1} \cdot \frac{6.02 \cdot 10^{23} \text{ photons}}{1 \text{ mol}} \cdot \frac{2.984 \cdot 10^{-25} \text{ J}}{1 \text{ photon}} = 0.233 \text{ J}$$

- i. Part b requires us to use Avogadro's number in a conversion factor. Write the number or symbol associated with it here:  $N_A = 6.02 \cdot 10^{23}$ . (0.5 pt)

11. An atom of a particular element has a volume of  $4.56 \cdot 10^{-24} \text{ nm}^3$ . What is this volume in cubic inches, providing your answer with the correct number of significant figures.  $1 \text{ m} = 3.28 \text{ ft}$   $1 \text{ ft} = 12 \text{ in}$  (3 pt)

$$\frac{4.56 \cdot 10^{-24} \text{ nm}^3}{1} \cdot \left( \frac{10^{-9} \text{ m}}{1 \text{ nm}} \right)^3 \cdot \left( \frac{3.28 \text{ ft}}{1 \text{ m}} \right)^3 \cdot \frac{12^3 \text{ in}^3}{1^3 \text{ ft}^3} = 2.78 \cdot 10^{-24} \text{ in}^3$$

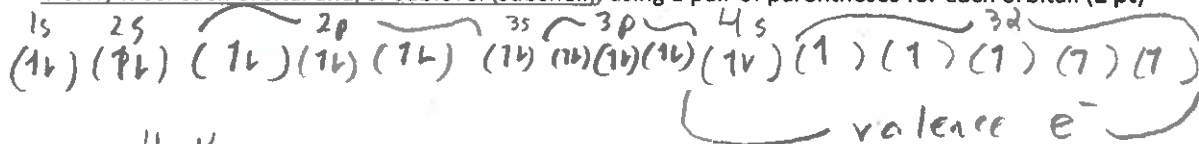
- a. Write a conversion factor (fraction or ratio) for the above question. (1 pt)  $\frac{1 \text{ m}}{3.28 \text{ ft}}$

12. Write the abbreviated electron configuration (noble gas configuration) for Manganese (Mn). (2 pt)

25 e<sup>-</sup>

- a. Given either the full orbital diagram, or the orbital diagram for the valence electrons of this element.

Clearly label each orbital and/or sublevel (subshell), using a pair of parentheses for each orbital. (2 pt)



- i. Hund's Rule should be followed when filling orbital diagrams. Fill in the blank and briefly describe this rule below. (Note getting the above part correct earns you this pt) (1 pt)

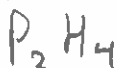
→ in a sublevel, e<sup>-</sup> will be unpaired w/ same spin (pointing same way)

13. What is the ionic charge when each element below is a monatomic ion, and what noble gas are all these ions, except one, isoelectronic with? Write the charges clearly for each and identify the noble gas. (2 pt)

Rb <sup>+</sup> Al <sup>3+</sup> As <sup>3-</sup> Isoelectronic noble gas = Kr

14. Provide the chemical formulas for the IUPAC names below. (3 pt)

Diphosphorus tetrahydride



Rubidium oxide



Iron (III) Sulfate



15. Identify the following elements as metals (M) or nonmetals (NM). (2 pt)

He

NM

Pb

M

Li

M

H

NM

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## Show your work

16. Identify whether the following have covalent (C), ionic (I) or both types (B) of bonding. (2 pt)

- a. NaCl      b. HCl      c. NH<sub>4</sub>Br      d. N<sub>2</sub>O<sub>4</sub>

I

C

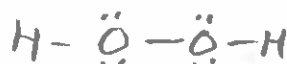
B

C

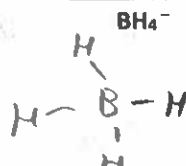
17. Draw the Lewis structures for H<sub>2</sub>O<sub>2</sub> and BH<sub>4</sub><sup>-</sup>. Use dots for unpaired (lone pair) electrons. Identifying the total number of electrons for each is a good move. Note one of these has two central atoms. (4 pt)

$2 \times 1 + 6 \times 2 = 14 \text{ valence } e^-$

H<sub>2</sub>O<sub>2</sub>



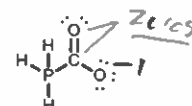
BH<sub>4</sub><sup>-</sup>



$3 + 4 \times 1 - (-1) = 8 \text{ val } e^-$

18. For the nonhydrogen atoms in this structure, clearly identify all formal charges. (2 pt)

P = +1



19. Briefly describe the use of VSEPR theory, and use this to identify the molecular or electron pair geometries of the P & C in the given structure. Note, saying VSEPR tells us geometry is not enough. (2 pt)

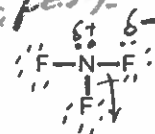
Geometry of P: tetrahedral

Geometry of C: trigonal planar

*e<sup>-</sup> pair around a atom repel other pairs & want to be far apart*

20. To the structure of NF<sub>3</sub> provided, add in the correct # of lone pairs as dots to give each atom a full octet. Add one bond dipole using δ<sup>+</sup> & δ<sup>-</sup>. Then identify whether this is a polar or nonpolar molecule. (3 pt)

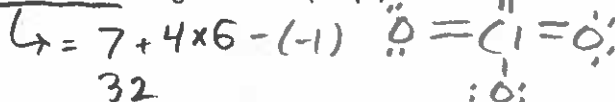
$5 + 3 \times 7 = 26 \text{ val } e^-$



a. Are all atoms in the molecule above in the same 2D plane? Yes or No. (0.5 pt)

b. What is the F - N - F bond angle? (0.5 pt) less than 109.5°

21. Draw the Lewis structure of ClO<sub>4</sub><sup>-</sup> where the central chlorine has a formal charge = zero. Identifying the total number of valence electrons is a good move. (2.5 pt)



*if all C-O bonds right, -1.5*

a. Which of the elements in the above formula can be hypervalent (have an expanded octet)? Circle it. (0.5 pt)

O

Cl

Neither can be hypervalent

22. The pressure of a sample of gas is measured at sea level (P<sub>atm</sub> = 760 mm Hg) with an open-end Hg (mercury) manometer (shown here). If h = 35 cm, determine the pressure of the gas in mm Hg. (2 pt)

h = 350 mm

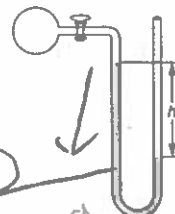
$P_{atm} = P_{Hg} + P_{gas} \rightarrow P_{gas} = P_{atm} - P_{Hg} = 760 - 350 = 410 \text{ mm Hg}$

a. What has greater pressure, the atmosphere or the gas? Circle it (1 pt)

b. Identify where in the manometer the mercury is. (1 pt)

c. How would h change if the gas was heated, while P<sub>atm</sub> was held constant? Would h increase, decrease, or stay constant. Circle it and briefly explain your answer. (1 pt)

*heated = higher gas T, More pressure, pushes Hg*

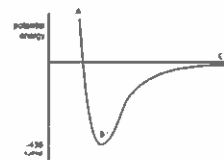


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Show your work

23. Forming chemical bonds is an **endothermic/exothermic** process, **releasing/absorbing/requiring** energy. Circle the correct bolded term in each pair. (2 pt)



- a. Using the letters on the given diagram of distance between atoms vs. energy, identify the process for forming a chemical bond. i.e. an answer could be A → D if D was also a point. (2 pt)

24. What is the volume (in L) of 0.350 kg propane gas ( $C_3H_8$ ) at a temperature of  $35^\circ C$  and a pressure of 1.05 atm. (4 pt)

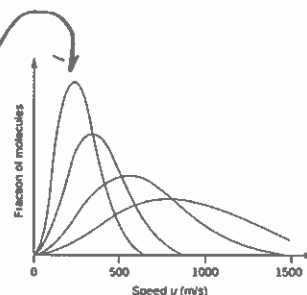
Handwritten work for Question 24:

$$PV = nRT$$

$$V = \frac{nRT}{P} = \frac{7.954 \text{ mol} \cdot 0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}} \cdot 308 \text{ K}}{1.05 \text{ atm}} = 192 \text{ L}$$

Additional notes:  $350 \text{ g } C_3H_8$ ,  $MM = (3 \times 12 + 8 \times 1) = 44 \text{ g/mol}$ ,  $7.954 \text{ mol}$ ,  $308 \text{ K}$ .

25. For a given temperature, identify the curve that corresponds to the **heaviest gas** (lowest MM). (1.5 pt)



- a. As temperature increases, what happens to the average kinetic energy for a gas? Briefly explain your answer. (1.5 pt)

as  $T \uparrow$ ,  $KE \uparrow$ , since thermal energy is energy of movement.

26. Balanced the reaction for the formation of  $NH_3$ :  $1 N_2 + 3 H_2 \rightarrow 2 NH_3$ . (1.5 pt)

- a. If 56 g of  $N_2$  gas reacts with an excess amount of  $H_2$  gas, and 6.0 g of  $NH_3$  forms, calculate the percent yield. Doing so requires you to calculate the **actual yield** or **theoretical yield**. Circle the correct term & calculate the percent. (3 pt)

Handwritten work for Question 26a:

$$\frac{56 \text{ g } N_2}{1} \cdot \frac{1 \text{ mol } N_2}{(14 \times 2) \text{ g } N_2} \cdot \frac{2 \text{ mol } NH_3}{1 \text{ mol } N_2} \cdot \frac{(14 + 3 \times 1) \text{ g } NH_3}{1 \text{ mol } NH_3} = 68 \text{ g } NH_3$$

Percent yield calculation:

$$\% \text{ Yield} = \frac{\text{Actual}}{\text{Theoretical}} \cdot 100\% = \frac{6.0}{68} \cdot 100\% = 8.82\%$$

- a. Describe how the idea of conservation of mass is **followed** or **broken** by actual, theoretical, and percent yields. (2 pt)

Actual Yield - if all the mass perfectly transferred into product.

Theoretical Yield - Does not account for the atoms becoming side products, but the atoms are still conserved in them!

- b. What phase of matter (s, l, g, or aq) is  $H_2$  and iron (Fe) commonly observed as? (1.5 pt)

$H_2$  (g) (wrong -1)

Fe (s)

- c. Is the main reaction above, an acid-base, precipitation, or redox reaction? Circle it and briefly explain your answer. (2 pt)

Oxidation states change!

Show your work

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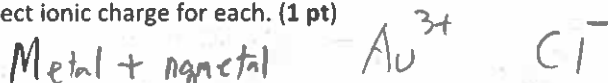
Show your work

27. Clearly identify the oxidation number (state) for each element below. Note, if the same element is given twice, it may have different oxidation numbers in different species. (3.5 pt)

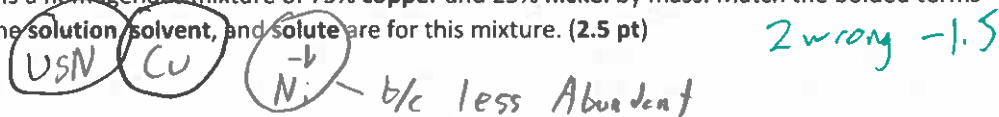


- a. Oxidation numbers are like \_\_\_\_\_. Circle the phrase below that goes in the blank. (0.5 pt)
- Electronegativity      Ionization Energy      Ionic Charge      Energy Transfer

- b. Other than  $\text{Na}_2\text{O}_2$ , another of the above species is an ionic compound. Write its cation and anion with the correct ionic charge for each. (1 pt)



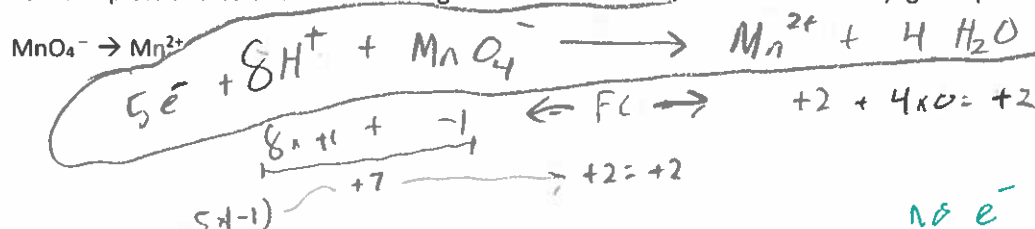
28. A US nickel (USN) is a homogenous mixture of 75% copper and 25% nickel by mass. Match the bolded terms to identify what the solution, solvent, and solute are for this mixture. (2.5 pt)



- a. Give an example of a heterogeneous mixture and explain why it is so. (1.5 pt)

→ A mixture that Looks Different throughout.  
Oil + Vinegar, etc.

29. Complete and balance the following half-reaction in acidic solution. You may ignore phases. (3 pt)



- a. Basic solution means which of the following may be present in your half-reaction:  $\text{H}^+$ ,  $\text{OH}^-$ , or  $\text{H}_2\text{O}$ . Circle all that may show up in a basic solution. (1 pt)

30. Complete and balance the following half-reaction. You may ignore phases. (2 pt)



- a. Identify whether this is a reduction or oxidation half-reaction. You must have a reasonable attempt in the main part to be able to earn this point. (1 pt)

31. In a balanced chemical equation or half-equation, Atoms & Net Charge must be balanced. Fill in the blanks with the correct terms/phrases. (1 pt)

32. A species that is reduced gains electrons. While a species that is an acid loses protons, or produces Hydronium. Lastly, isotopes have the same number of protons and different numbers of neutrons. Correctly fill in the blanks with the below words. A word may be used multiple times. Note if you would rather use a different word, you may, but the words below will work. (3 pt)

| Hydroxide(s) | Proton(s) | Electron(s) | Neutron(s)  | Hydronium(s) |
|--------------|-----------|-------------|-------------|--------------|
| Loses        | Gains     | Transmutes  | Capitalizes |              |

Key

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33. 0.616 mol of vitamin C was determined to be 0.0035 g vitamin C. What is the molar mass (molecular weight) of vitamin C? (2 pt)

$$MM = \frac{g}{mol} = \frac{0.616g}{0.0035mol} = 176g/mol$$

Flipped -15

- a. A 1 g sample of vitamin C was subjected to combustion analysis and determined to be composed only of C, H, and O. 1.5 g of CO<sub>2</sub> and 0.41 g H<sub>2</sub>O was generated. Determine the empirical formula of vitamin C. (2 pt)

total g C 1 - 0.41 - 0.0456 = 0.5444 g O

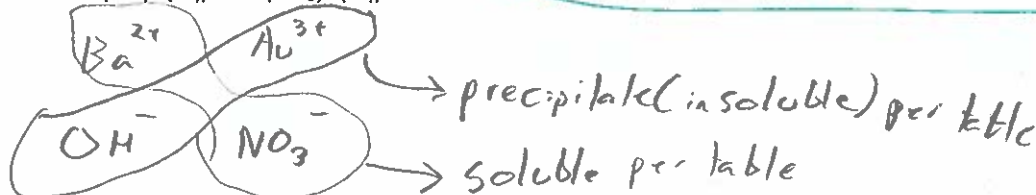
$$0.034mol\ O$$

$$1.5g\ CO_2 \cdot \frac{1mol\ CO_2}{44g\ CO_2} \cdot \frac{1mol\ C}{1mol\ CO_2} \cdot \frac{12g\ C}{1mol\ C} = 0.41g\ C \quad [or\ 0.034mol\ C]$$

$$0.41g\ H_2O \cdot \frac{1mol\ H_2O}{18g\ H_2O} \cdot \frac{2mol\ H}{1mol\ H_2O} \cdot \frac{1g\ H}{1mol\ H} = 0.0456g\ H$$

$$(C_3H_4O_3) \leftarrow (C_{1.34}H_{1.34}O_1) \times 3 \leftarrow \frac{0.034\ H}{0.034} \frac{0.0456}{0.034} \frac{0.034}{0.034}$$

34. Complete the balanced net ionic equation for the precipitation reaction below. If no reaction occurs, write NR after the reaction arrow. Be sure to include all proper phases for all species in the reaction. You should show some work using the solubility tables on the equation sheet to explain your answer. (3 pt)



- a. Provide the IUPAC names for each of the starting ionic compounds in the above problem. (2 pt)



Gold (III) Nitrate

35. Write 6 pairs of polyatomic ions, of correct formula & charges and respective names that you were asked to memorize. You may use ones present in this exam. (2 pt)

Slides

36. The density of a certain gaseous compound is 3.93 g/L at STP (273 K & 1 atm). Calculate the molar mass of this compound. (3 pt)

Can solve via PV=nRT but

1 mol gas @ STP = 22.4 L

$$22.4L \cdot 3.93g/L = 88.032g$$

$$MM = \frac{88.032g}{1mol} = 88.03g/mol$$

- a. Assume you have 1 mol of the compound. Fill in the blank with a number and a unit. Getting the main part right earns you this pt. (1 pt)

- b. Write the equation for the ideal gas law. (2 pt)

PV=nRT